

DIGITAL EGYPT PIONEERS INITIATIVE

Data Analytics Track

UK ROAD ACCIDENTS DATA ANALYSIS

Excel • SQL Server • Power BI

Project Title	UK Road Accidents Data Analysis
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1. Executive Summary

1.1 Problem Statement

Road traffic accidents remain a persistent public health and economic burden in the United Kingdom, resulting in thousands of casualties and significant socioeconomic cost every year. Despite continuous investment in road infrastructure and traffic regulation, accident frequency and severity are influenced by a complex interaction of factors including weather conditions, road surface state, lighting, junction design, speed limits, and vehicle type. Without a structured, data-driven understanding of these interactions, road safety authorities are limited in their ability to prioritize interventions and allocate resources effectively.

This project addresses that gap by transforming a raw, large-scale government road-accident dataset into actionable intelligence through a complete data analytics pipeline from data engineering to interactive visual storytelling.

1.2 Primary Objectives

- Consolidate and clean a real-world UK road accident dataset covering the 2021–2022 period, comprising 307,973 individual accident records.
- Design a relational SQL database schema to structure the data for efficient querying, aggregation, and analytical reporting.
- Perform exploratory data analysis (EDA) to identify the temporal, environmental, and structural factors most strongly associated with accident frequency and severity.
- Build interactive Power BI and Tableau dashboards that allow stakeholders to explore accident patterns dynamically across time, geography, and category.
- Translate analytical findings into concrete, actionable recommendations that can support road-safety policymaking and resource allocation.

1.3 High-Impact Insights

The analysis of the full dataset (307,973 accidents, 417,883 total casualties, 563,302 vehicles involved) surfaced several high-impact findings that directly inform road-safety strategy:

- Severity distribution is heavily skewed toward less severe outcomes: 85.5% of accidents are classified as Slight, 13.2% as Serious, and 1.3% as Fatal yet Fatal accidents alone account for thousands of preventable deaths annually and warrant disproportionate policy attention.
- Single carriageways account for 74.9% of all accidents and, more critically, 74.0% of fatal casualties in 2022 making them the single highest-priority road type for safety intervention.

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- Fine weather with no high winds is associated with 79.4% of all accidents, indicating that driver behavior and road design — not adverse weather — are the dominant contributing factors, countering the common assumption that bad weather is the primary accident driver.
 - Friday is consistently the highest-risk day of the week, while Sunday records the fewest accidents, a pattern consistent with elevated end-of-week traffic volume and commuter fatigue.
 - Urban areas host 64.5% of accidents by volume, but rural roads and higher speed-limit zones (60–70 mph) show a disproportionately higher severity rate, indicating that rural single-carriageway roads are a critical target for targeted safety campaigns.
 - Cars are involved in 77.9% of all recorded accidents and 2,302 of the 2,855 fatal casualties recorded in 2022 alone, reflecting their overwhelming share of UK road traffic rather than a disproportionate risk factor per vehicle.

2. Project Workflow & Methodology

The project followed a standard end-to-end data analytics lifecycle, moving the dataset through six discrete stages. Each stage used the tool best suited to its purpose, reflecting real-world data analytics team practices where Excel, SQL, and BI platforms operate as complementary layers rather than substitutes for one another.

2.1 End-to-End Data Pipeline

Stage 1 — Data Collection

The raw dataset was sourced as a structured CSV export containing 307,973 accident records spanning the 2021–2022 reporting years, with 19 attributes per record covering accident metadata, environmental conditions, location, and vehicle involvement.

Stage 2 — Data Cleaning

Initial data quality assessment was performed in Excel, including detection of missing values, inconsistent categorical labels, duplicate accident indices, and outlier casualty/vehicle counts. Standardization rules were applied to ensure categorical fields (severity, weather, road type, etc.) used consistent naming conventions throughout.

Stage 3 — Transformation

Cleaned data was transformed into an analysis-ready structure: date fields were parsed into year/month/day components, time fields were standardized into 24-hour format for hour-of-day analysis, and categorical fields were validated against a fixed domain of accepted values to support reliable aggregation in SQL.

Stage 4 — Exploratory Data Analysis (EDA)

SQL Server was used to run systematic aggregation queries across every dimension of the dataset — severity, time, weather, road type, geography, and vehicle type — establishing the core statistical foundation referenced throughout Section 5.

Stage 5 — Modeling

A relational schema was designed around a central accident fact table, with the analytical model supporting star-schema-style aggregation across severity, location, time, and environmental dimensions (see Section 4.2).

Stage 6 — Visualization

Power BI and Tableau were used in parallel to build complementary interactive dashboards, allowing the team to compare visual storytelling approaches and deliver insights suited to different stakeholder audiences (see Section 6).

2.2 Tool Contribution Matrix

Tool	Pipeline Stage	Primary Contribution
Excel	Cleaning & Initial EDA	Missing-value audit, category standardization, pivot-table-based first-pass exploration
SQL Server	Modeling & Aggregation	Schema design, relational storage, aggregation queries, KPI computation
Power BI	Visualization	Interactive dashboards, drill-down filtering, YoY comparison, KPI cards

3. Data Preparation & SQL Database Architecture

3.1 Data Cleaning & Missing Value Handling

The dataset was audited column-by-column for completeness and consistency prior to modeling. The following procedures were applied:

- Null-value audit: each of the 19 fields was checked for missing values using COUNT(*) minus COUNT(column) comparisons in SQL, allowing a quick per-column completeness score without complex conditional logic.
- Duplicate detection: the accident_index field (the natural primary key) was grouped and checked for records occurring more than once, confirming record-level uniqueness.
- Categorical standardization: fields such as accident_severity, weather_conditions, and road_surface_conditions were checked with SELECT DISTINCT to catch inconsistent labels (e.g., casing or spacing variants) before aggregation.
- Type validation: numeric fields (number_of_casualties, number_of_vehicles, speed_limit) were validated to ensure no non-numeric or negative values existed.
- Date/time parsing: the accident_date field was cast to a proper DATE type and the time field to a TIME type to enable native SQL Server date-part functions (YEAR, MONTH, DATEPART) for temporal analysis.

3.2 Core SQL Queries for Aggregation & Extraction

Below are representative queries from the full analytical query set used to extract the KPIs discussed in Section 5. All queries were written and tested in Microsoft SQL Server .

Overall KPI Summary

```
SELECT
    COUNT(*)                AS total_accidents,
    SUM(number_of_casualties) AS total_casualties,
    SUM(number_of_vehicles)  AS total_vehicles_involved,
    SUM(number_of_casualties) * 1.0 / COUNT(*) AS avg_casualties_per_accident
FROM road_accident;
```

Severity Distribution

```
SELECT accident_severity, COUNT(*) AS accident_count
FROM road_accident
GROUP BY accident_severity
ORDER BY accident_count DESC;
```

Monthly Accident Trend

```
SELECT YEAR(accident_date) AS yr, MONTH(accident_date) AS mth,  
       COUNT(*) AS accident_count, SUM(number_of_casualties) AS total_casualties  
FROM road_accident  
GROUP BY YEAR(accident_date), MONTH(accident_date)  
ORDER BY yr, mth;
```

Fatal Casualties by Road Type (Highest-Risk Infrastructure)

```
SELECT road_type, SUM(number_of_casualties) AS fatal_casualties  
FROM road_accident  
WHERE accident_severity = 'Fatal'  
GROUP BY road_type  
ORDER BY fatal_casualties DESC;
```

Top-Risk Local Authorities

```
SELECT TOP 10 local_authority, COUNT(*) AS accident_count,  
       SUM(number_of_casualties) AS total_casualties  
FROM road_accident  
GROUP BY local_authority  
ORDER BY accident_count DESC;
```


4. Exploratory Data Analysis (EDA) & Key Insights

4.1 Core KPIs

307,973 TOTAL ACCIDENTS	417,883 TOTAL CASUALTIES	563,302 VEHICLES INVOLVED	1.36 AVG CASUALTIES / ACCIDENT
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4.2 Severity Analysis

Across the full 2021–2022 period, accident severity is heavily concentrated in the lower-impact category, but the absolute scale of Fatal and Serious outcomes remains substantial:

Severity	Accident Count	% of Total
Slight	263,280	85.49%
Serious	40,740	13.23%
Fatal	3,953	1.28%

Table 4.1 — Accident severity distribution, full dataset

Year-over-year, both accident volume and severity improved: total accidents fell from 163,554 in 2021 to 144,419 in 2022 (a decline of 11.7%), while Fatal accidents dropped from 2,404 to 1,549 (a 35.6% reduction) — a strong positive signal, though still representing over 2,850 fatal casualties in 2022 alone.

4.3 Temporal Patterns

Day of Week

Day	Accident Count
Friday	50,529
Tuesday	46,386
Wednesday	46,381
Thursday	45,649
Monday	43,918
Saturday	41,566
Sunday	33,544

Table 4.2 — Accidents by day of week

Friday consistently records the highest accident volume, likely reflecting peak weekly traffic density combined with end-of-week commuter and leisure travel. Sunday is the lowest-risk day, roughly 34% below Friday's volume.

Peak Hours

Hourly analysis of accident timestamps confirms two distinct peaks aligned with the UK commuter schedule: a morning peak around 08:00, and a more pronounced afternoon/evening peak between 15:00 and 18:00, consistent with school-run and homebound-commute traffic. This dual-peak structure is a critical input for time-targeted traffic enforcement and public safety campaigns.

4.4 Environmental Conditions

Weather

Weather Condition	Accident Count	% of Total
Fine, no high winds	244,496	79.39%
Raining, no high winds	37,841	12.29%
Other	8,802	2.86%
Snowing, no high winds	4,839	1.57%
Fog or mist	4,783	1.55%
Raining + high winds	3,526	1.14%
Fine + high winds	3,148	1.02%
Snowing + high winds	538	0.17%

Table 4.3 — Accidents by weather condition

Counter to a common public assumption, adverse weather is not the leading contributor to accident volume — nearly four out of five accidents occur in fine weather with no high winds. This suggests that driver behavior, traffic density, and road/junction design are more influential risk factors than weather itself, and that safety interventions should not be weather-contingent alone.

Road Surface & Lighting

Road Surface	Accident Count	% of Total
Dry	208,967	67.86%
Wet or damp	81,796	26.56%
Frost or ice	12,078	3.92%
Snow	4,758	1.55%

Road Surface	Accident Count	% of Total
Flood (over 3cm deep)	374	0.12%

Table 4.4 — Accidents by road surface condition

A parallel pattern holds for lighting: 73.8% of accidents occur in daylight, with the remaining share distributed across various darkness conditions. Combined with the weather findings, this reinforces that the majority of accidents happen under normal, non-adverse driving conditions — placing greater analytical weight on structural and behavioral factors (road type, junction design, speed) as primary levers for risk reduction.

4.5 Road Infrastructure & Speed

Road Type	Accident Count	% of Total
Single carriageway	230,612	74.88%
Dual carriageway	45,467	14.76%
Roundabout	20,929	6.80%
One way street	6,197	2.01%
Slip road	4,768	1.55%

Table 4.5 — Accidents by road type

Single carriageways are both the most common accident location and, critically, the site of the vast majority of fatal outcomes — accounting for 74.0% of all fatal casualties recorded in 2022, compared to 23.7% for dual carriageways. This makes single-carriageway roads the single highest-priority target for infrastructure-level safety investment (e.g., median barriers, speed reduction, improved signage).

Speed limit analysis shows that 30 mph zones — typically urban residential and arterial roads — account for 65.0% of all accidents by sheer traffic volume, while higher-speed rural roads (60–70 mph) show a smaller share of total accidents but a disproportionately higher severity rate per incident, consistent with the physics of higher-speed collisions.

4.6 Geography & Urban vs. Rural Risk

Area Type	Accident Count	% of Total
Urban	198,532	64.46%
Rural	109,441	35.54%

Table 4.6 — Accidents by urban/rural classification

While urban areas host nearly two-thirds of all accidents by volume, rural roads — which are more likely to be single carriageways with higher speed limits and reduced lighting — carry a higher risk of severe or fatal outcomes per accident. This urban-volume / rural-severity split is a recurring pattern in national road safety statistics and is confirmed in this dataset.

At the local authority level, Birmingham (6,165 accidents), Leeds (4,140), and Manchester (3,132) are the top three geographic hotspots by volume — unsurprising given their status as the UK's largest metropolitan areas, but still valuable for prioritizing local enforcement and infrastructure review.

4.7 Vehicle Type Involvement

Vehicle Type	Accident Count	% of Total
Car	239,794	77.86%
Van / Goods ≤3.5t	15,695	5.10%
Motorcycle >500cc	11,226	3.65%
Bus or coach	8,686	2.82%
Motorcycle ≤125cc	6,852	2.22%
Goods ≥7.5t	6,532	2.12%
Taxi / Private hire	5,543	1.80%

Table 4.7 — Accidents by primary vehicle type (top 7)

Cars dominate both accident involvement and fatal casualty counts (2,302 of 2,855 fatal casualties in 2022), which is proportional to their overwhelming share of UK road traffic rather than an indicator of elevated per-vehicle risk. Motorcycles, while a small share of total accidents, warrant separate attention in the recommendations given the well-documented disproportionate injury severity associated with two-wheeled vehicles.

5. Dashboards & Interactive Visualizations

Two complementary dashboards were developed, one in Power BI and one in excel allowing the team to compare BI platforms while delivering the same core analytical narrative through different visual design languages. Both dashboards follow a dark-themed, high-contrast layout optimized for KPI scanning and geographic pattern recognition.

5.1 Power BI Dashboard

The Power BI dashboard is structured around five KPI cards (Total Accidents, Total Casualties, Fatal Casualties, Serious Casualties, Slight Casualties), each paired with a monthly trend sparkline and a year-over-year percentage change indicator. Supporting visuals include:

- A donut chart breaking down fatal casualties by weather condition and road surface condition, enabling rapid identification of the environmental conditions most associated with severe outcomes.
- A horizontal bar chart ranking fatal casualties by road type, directly surfacing single carriageways as the dominant risk category.
- An icon-based summary of fatal casualties by vehicle type, giving stakeholders an at-a-glance comparison across cars, motorcycles, buses, and goods vehicles.
- A geographic map plotting individual fatal casualty locations across the UK, revealing regional density clusters concentrated around major metropolitan areas.
- Interactive slicers for Current Year, Previous Year, and Accident Severity, allowing users to dynamically compare periods and isolate specific severity levels across every visual on the page.

5.3 Visual Storytelling & Interactivity Design Principles

Both dashboards were designed around three core principles: (1) lead with KPIs so stakeholders immediately grasp scale before detail, (2) pair every summary metric with a comparative reference — year-over-year change or percentage-of-total so numbers are never presented without context, and (3) make every chart a filter, so that a single click on any category (e.g., a road type or weather condition) propagates across the entire dashboard, supporting exploratory, hypothesis-driven analysis for non-technical stakeholders such as policymakers.

6. Strategic Recommendations & Conclusion

6.1 Actionable Recommendations

1. Prioritize single-carriageway safety upgrades

Given that single carriageways account for nearly 75% of all accidents and fatal casualties, infrastructure investment (median separation, junction redesign, targeted speed reduction) should be concentrated on this road type ahead of dual carriageways or motorways.

2. Shift enforcement focus away from weather-based assumptions

Since the vast majority of accidents occur in fine weather and daylight, traffic enforcement and public awareness campaigns should focus on driver behavior (speeding, distraction, junction discipline) rather than being weighted toward adverse-weather messaging alone.

3. Target Friday and peak commute-hour enforcement

Increased visible policing and variable speed enforcement during Friday traffic and the 15:00–18:00 evening peak window would address the periods of highest observed accident density.

4. Address the rural severity gap

While urban roads carry higher accident volume, rural single-carriageway roads with higher speed limits show a disproportionate severity rate. Rural-specific interventions — improved lighting, hazard signage, and speed-limit review — should be prioritized even where raw accident counts are lower than urban areas.

5. Expand motorcycle-specific safety initiatives

Given the well-established disproportionate injury severity for motorcyclists relative to their traffic share, targeted awareness campaigns and infrastructure considerations (e.g., junction visibility) for motorcycles over 500cc are recommended.

6. Continue year-over-year monitoring

The observed 11.7% reduction in total accidents and 35.6% reduction in fatal accidents between 2021 and 2022 indicates existing interventions are showing measurable impact. Sustained, granular monitoring ideally through a live-refreshed version of the dashboards built in this project is recommended to track whether this positive trend continues and to detect any reversal early.

6.2 Conclusion

This project demonstrates a complete, professional-grade data analytics workflow from raw CSV ingestion through SQL-based modeling and aggregation, to dual-platform interactive dashboard delivery applied to a real-world, high-stakes public safety dataset of nearly 308,000 records. The resulting insights move beyond descriptive statistics to identify specific, prioritized, and actionable levers for road safety

improvement: infrastructure investment in single carriageways, behavior-focused (not weather-focused) enforcement, temporal targeting of peak-risk periods, and continued rural-urban severity monitoring.

The integration of Excel, SQL Server, and Power BI across the pipeline reflects the multi-tool fluency expected of a professional data analyst, while the recommendations translate technical findings into a form directly usable by non-technical policymakers and road safety authorities fulfilling the project's core objective of supporting data-driven decision-making for public road safety.

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